N3_Gigs_Calculations

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USING INTERGROWTH-21 ST EXTENSION TO GET SIZE FOR GESTATIONAL AGE - JN This code uses the INTERGROWTH-21st macro available at https://github.com/ropensci/gigs (for Stata codes see: https://github.com/lshtm-gigs) to calculate newborn size for gestational age.

The standard IG-21 contains the gestational age range from 24+0 to 42+6 weeks, but their extension macro allows for the extrapolation from 22+0 to 44+6 weeks.

These codes use the extension macro, allowing for a larger GA range. It contains the following IG-21st outputs: 1. Birthweight: 1.1 Centile: birthweight_centile 1.2 Z-score: birthweight_zscore 1.3 SGA, LGA, and sfga_category (SGA, AGA, LGA) 2. Birth length: 2.1 Centile: blen_centile 2.2 Z-score: blen_zscore 2. Birth head circumference: 2.1 Centile: bhc_centile 2.2 Z-score: bhc_zscore

Outlier SD values are also calculated for each z-score.

Reading in N3 newborn weight dataset.

```
n3bwt_data <- read_dta("/Users/angelakc/Library/CloudStorage/OneDrive-JohnsHopkins/Intergrowth 21_2026/
```

Check how Gigs file works and the syntax.

```
?gigs
help(package = "gigs")
```

Recode sex.

```
# Create sex_mf variable: 1 = "M" (male), 2 = "F" (female)
n3bwt_data$sex_mf <- ifelse(n3bwt_data$sex_new == "Male", "M", "F")
```

Obtain birthweight in kg.

```
n3bwt_data$bwt <- n3bwt_data$bwt_comb/1000
```

Classify size for GA.

```r
# Create classified dataset
n3bwt_classified <- n3bwt_data

# Convert birth weight values to centiles using IG-NBS extension
n3bwt_classified$birthweight_centile <- value2centile(
  y = n3bwt_classified$bwt,           # weight in kg
  x = n3bwt_classified$ga,            # gestational age in days
  sex = n3bwt_classified$sex_mf,      # sex as "M"/"F"
  family = "ig_nbs_ext",              # IG-NBS extension standards
  acronym = "wfga"                    # weight-for-gestational-age
) |>
  round(digits = 3)


# Create individual binary indicators (SGA, LGA and AGA)
n3bwt_classified <- n3bwt_classified %>%
  mutate(
    sga = as.numeric(birthweight_centile < 0.10),
    lga = as.numeric(birthweight_centile > 0.90),
    aga = as.numeric(between(birthweight_centile, 0.10, 0.90))
  )

# Create a categorical variable
n3bwt_classified$sfga_category <- case_when(
  is.na(n3bwt_classified$birthweight_centile) ~ NA_character_,
  n3bwt_classified$birthweight_centile < 0.10 ~ "SGA",
  n3bwt_classified$birthweight_centile > 0.90 ~ "LGA",
  TRUE ~ "AGA"
)

# Add labels
n3bwt_classified$sfga_category <- factor(n3bwt_classified$sfga_category,
                                         levels = c("SGA", "AGA", "LGA"))
```

Calculate Z-scores birthweight

```r
# Define z-score cutoffs (including 0)
z_cutoffs <- c(-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6)

# Create labels for the categories
z_labels <- c("< -6 SD", "-6 to -5 SD", "-5 to -4 SD", "-4 to -3 SD",
              "-3 to -2 SD", "-2 to -1 SD", "-1 to 0 SD", "0 to +1 SD",
              "+1 to +2 SD", "+2 to +3 SD", "+3 to +4 SD", "+4 to +5 SD",
              "+5 to +6 SD", "> +6 SD")

# Function to categorize z-scores
categorize_zscore <- function(zscore) {
  cut(zscore,
      breaks = c(-Inf, z_cutoffs, Inf),
      labels = z_labels,
      include.lowest = TRUE)
```

```r
}

# Convert birth weight values to z-scores using IG-NBS extension
n3bwt_classified$birthweight_zscore <- value2zscore(
  y = n3bwt_classified$bwt,           # weight in kg
  x = n3bwt_classified$ga,            # gestational age in days
  sex = n3bwt_classified$sex_mf,      # sex as "M"/"F"
  family = "ig_nbs_ext",              # IG-NBS extension standards
  acronym = "wfga"                    # weight-for-gestational-age
) |>
  round(digits = 3)

# Create birthweight categories
n3bwt_classified$birthweight_category <- categorize_zscore(n3bwt_classified$birthweight_zscore)

# Create zscore_outlier variable for +/- 5 SD as numeric
n3bwt_classified$zscore_5sd_outlier <- as.numeric(
  n3bwt_classified$birthweight_zscore < -5 | n3bwt_classified$birthweight_zscore > 5
)
```

Birth length classification - by GIGS

```r
# Convert birth length values to centiles using IG-NBS extension
n3bwt_classified$blen_centile <- value2centile(
  y = n3bwt_classified$blth,          # length in cm
  x = n3bwt_classified$ga,            # gestational age in days
  sex = n3bwt_classified$sex_mf,      # sex as "M"/"F"
  family = "ig_nbs_ext",              # IG-NBS extension standards
  acronym = "lfga"                    # length-for-gestational-age
) |>
  round(digits = 3)

# Convert birth length values to z-scores using IG-NBS extension
n3bwt_classified$blen_zscore <- value2zscore(
  y = n3bwt_classified$blth,          # length in cm
  x = n3bwt_classified$ga,            # gestational age in days
  sex = n3bwt_classified$sex_mf,      # sex as "M"/"F"
  family = "ig_nbs_ext",              # IG-NBS extension standards
  acronym = "lfga"                    # length-for-gestational-age
) |>
  round(digits = 3)

# Create birth length categories
n3bwt_classified$blen_category <- categorize_zscore(n3bwt_classified$blen_zscore)

# Create zscore_outlier variable for +/- 5 SD as numeric
n3bwt_classified$blen_zscore_5sd_outlier <- as.numeric(
  n3bwt_classified$blen_zscore < -5 | n3bwt_classified$blen_zscore > 5
)

# Create zscore_6sd_outlier variable for +/- 6 SD as numeric
n3bwt_classified$blen_zscore_6sd_outlier <- as.numeric(
```

```r
  n3bwt_classified$blen_zscore < -6 | n3bwt_classified$blen_zscore > 6
)
```

Birth head circumference classification - by GIGS

```r
# Convert head circumference values to centiles using IG-NBS extension
n3bwt_classified$hc_centile <- value2centile(
  y = n3bwt_classified$bhc,           # head circumference in cm
  x = n3bwt_classified$ga,            # gestational age in days
  sex = n3bwt_classified$sex_mf,      # sex as "M"/"F"
  family = "ig_nbs_ext",              # IG-NBS extension standards
  acronym = "hcfga"                   # head circumference-for-gestational-age
) |>
  round(digits = 3)

# Convert head circumference values to z-scores using IG-NBS extension
n3bwt_classified$hc_zscore <- value2zscore(
  y = n3bwt_classified$bhc,           # head circumference in cm
  x = n3bwt_classified$ga,            # gestational age in days
  sex = n3bwt_classified$sex_mf,      # sex as "M"/"F"
  family = "ig_nbs_ext",              # IG-NBS extension standards
  acronym = "hcfga"                   # head circumference-for-gestational-age
) |>
  round(digits = 3)

# Create head circumference categories
n3bwt_classified$hc_category <- categorize_zscore(n3bwt_classified$hc_zscore)

# Create zscore_outlier variable for +/- 5 SD as numeric
n3bwt_classified$hc_zscore_5sd_outlier <- as.numeric(
  n3bwt_classified$hc_zscore < -5 | n3bwt_classified$hc_zscore > 5
)

# Create zscore_6sd_outlier variable for +/- 6 SD as numeric
n3bwt_classified$hc_zscore_6sd_outlier <- as.numeric(
  n3bwt_classified$hc_zscore < -6 | n3bwt_classified$hc_zscore > 6
)
```

```r
# Export the dataset as a Stata .dta file
write_dta(n3bwt_classified, "/Users/angelakc/Library/CloudStorage/OneDrive-JohnsHopkins/Intergrowth 21_
```